FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Systems of Particles

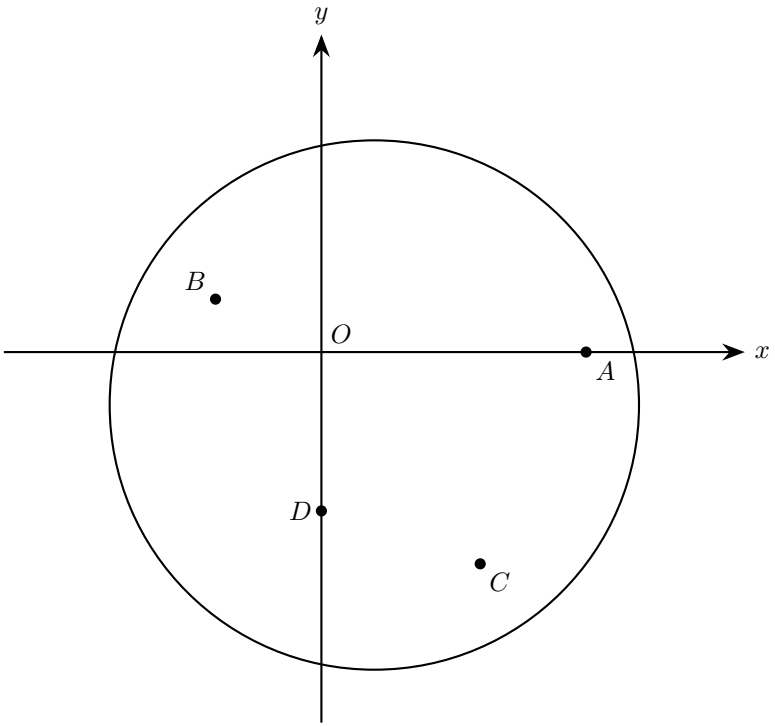
Worked solutions 9 questions · 59 marks

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*4 marks*	2
Question 2	*5 marks*	4
Question 3	*4 marks*	6
Question 4	*4 marks*	8
Question 5	*6 marks*	9
Question 6	*9 marks*	11
Question 7	*8 marks*	13
Question 8	*7 marks*	15
Question 9	*12 marks*	17



1. Four sensors are attached to a circular plate.

The plate is to be modelled as a uniform lamina and the four sensors as four particles.

The diagram shows the lamina, the four particles A, B, C and D, and the coordinate axes.

The lamina has mass 6 kg and its centre of mass is at the point $(1, -1)$.

Particle A has mass 4 kg and is at the point $(5, 0)$.
Particle B has mass 3 kg and is at the point $(-2, 1)$.
Particle C has mass 5 kg and is at the point $(3, -4)$.
Particle D has mass 2 kg and is at the point $(0, -3)$.

Find the coordinates of the centre of mass of the system of plate and sensors. **[4]**

Solution

Treat the uniform lamina as a particle of mass 6 kg acting at its centre of mass $(1, -1)$.

Let the centre of mass of the whole system be $(\bar{x}, \bar{y})$.

For particles of masses m_i at the points (x_i, y_i), the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = 6 + 4 + 3 + 5 + 2 = 20\,\text{kg}$$

For the x-coordinate:

$$\begin{aligned} 20\bar{x} &= 6(1) + 4(5) + 3(-2) + 5(3) + 2(0) \\ &= 6 + 20 - 6 + 15 + 0 \\ &= 35 \end{aligned}$$

Hence

$$\bar{x} = \frac{35}{20} = \frac{7}{4}$$

For the y-coordinate:

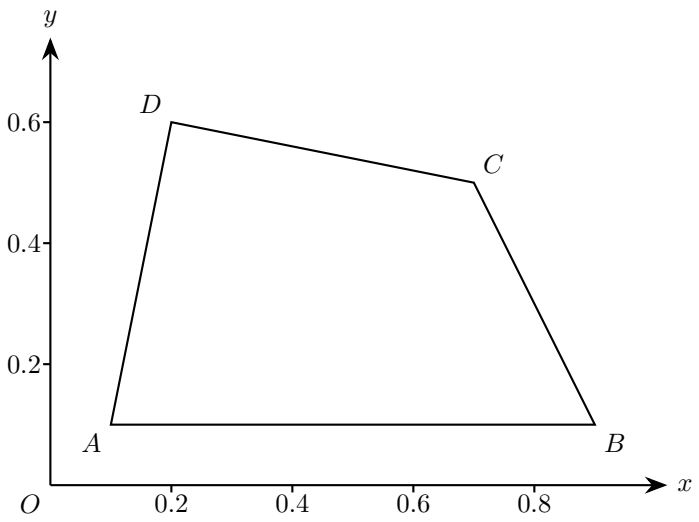
$$\begin{aligned} 20\bar{y} &= 6(-1) + 4(0) + 3(1) + 5(-4) + 2(-3) \\ &= -6 + 0 + 3 - 20 - 6 \\ &= -29 \end{aligned}$$

Hence

$$\bar{y} = -\frac{29}{20}$$

Therefore the centre of mass of the system is

$$\left(\frac{7}{4}, -\frac{29}{20}\right)$$



2. Particles A, B, C and D of masses 1 kg, 2 kg, 3 kg and 4 kg respectively are joined by light rigid rods to form the quadrilateral frame shown in the diagram. The particles are at the points $A(0.1, 0.1)$, $B(0.9, 0.1)$, $C(0.7, 0.5)$ and $D(0.2, 0.6)$, where coordinates are measured in metres.

G, the centre of mass of the frame, has coordinates $(\bar{x}, \bar{y})$.

(a) Show that $\bar{y} = 0.42$. Find $\bar{x}$. **[4]**

(b) Using your answer to part (a), explain why the frame cannot be in equilibrium in a horizontal plane if it is supported by a single peg touching the frame at one point. **[1]**

Solution

(a) For particles of masses m_i at the points (x_i, y_i), the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = 1 + 2 + 3 + 4 = 10\,\text{kg}$$

For the y-coordinate:

$$\begin{aligned} 10\bar{y} &= 1(0.1) + 2(0.1) + 3(0.5) + 4(0.6) \\ &= 0.1 + 0.2 + 1.5 + 2.4 = 4.2 \\ \bar{y} &= \frac{4.2}{10} = 0.42 \end{aligned}$$

So $\bar{y} = 0.42$, as required.

For the x-coordinate:

$$\begin{aligned} 10\bar{x} &= 1(0.1) + 2(0.9) + 3(0.7) + 4(0.2) \\ &= 0.1 + 1.8 + 2.1 + 0.8 = 4.8 \\ \bar{x} &= \frac{4.8}{10} = 0.48 \end{aligned}$$

Hence the centre of mass is

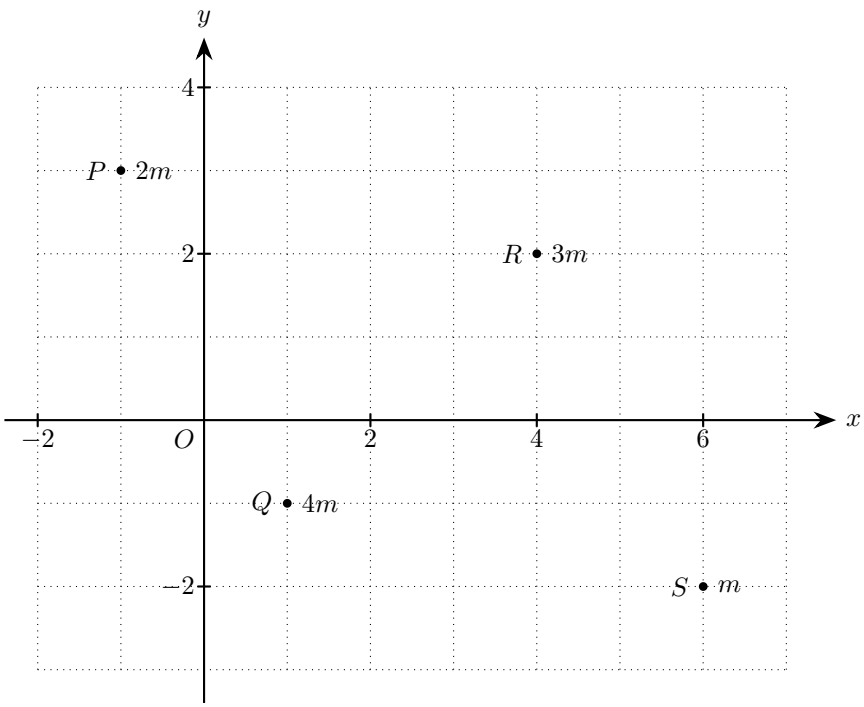
$$G = (\bar{x}, \bar{y}) = (0.48, 0.42)$$

(b) For equilibrium on a single peg, the peg must be vertically below the centre of mass.

So in the horizontal plane, the peg would have to touch the frame at $G = (0.48, 0.42)$.

But this point is not on the frame itself; it lies inside the quadrilateral, not on any of its sides.

Therefore the frame cannot be in equilibrium if it is supported by a single peg touching the frame at one point.



3. The diagram shows a system of four particles of masses $2m$, $4m$, $3m$ and m situated in the x-y plane at the points $P(-1, 3)$, $Q(1, -1)$, $R(4, 2)$ and $S(6, -2)$ respectively.

Find the coordinates of the centre of mass of the system. **[4]**

Solution

Let the centre of mass be $(\bar{x}, \bar{y})$.

For particles of masses m_i at the points (x_i, y_i), the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = 2m + 4m + 3m + m = 10m$$

For the x-coordinate:

$$M\bar{x} = 2m(-1) + 4m(1) + 3m(4) + m(6)$$

So

$$10m\bar{x} = -2m + 4m + 12m + 6m = 20m$$

Hence

$$\bar{x} = 2$$

For the y-coordinate:

$$M\bar{y} = 2m(3) + 4m(-1) + 3m(2) + m(-2)$$

So

$$10m\bar{y} = 6m - 4m + 6m - 2m = 6m$$

Hence

$$\bar{y} = \frac{3}{5}$$

Therefore the coordinates of the centre of mass are $(2, \frac{3}{5})$.

4. Four particles of mass m, $2m$, $4m$ and λm are placed at points with coordinates $(0,0)$, $(2,3)$, $(1,-1)$ and $(4,2)$ respectively. The centre of mass of the system is at $(2,k)$.

(a) Show that $\lambda = 3$. **[2]**

(b) Calculate the value of k. **[2]**

Solution

(a) For particles of masses m_i at the points (x_i, y_i), the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = m + 2m + 4m + \lambda m = (7 + \lambda)m$$

For the x-coordinate, with $\bar{x} = 2$:

$$M(2) = m(0) + 2m(2) + 4m(1) + \lambda m(4)$$

So

$$\begin{aligned} 2(7+\lambda)m &= 0 + 4m + 4m + 4\lambda m \\ 2(7+\lambda) &= 8 + 4\lambda \\ 14 + 2\lambda &= 8 + 4\lambda \\ 6 &= 2\lambda \\ \lambda &= 3 \end{aligned}$$

So $\lambda = 3$, as required.

(b) With $\lambda = 3$, the total mass is

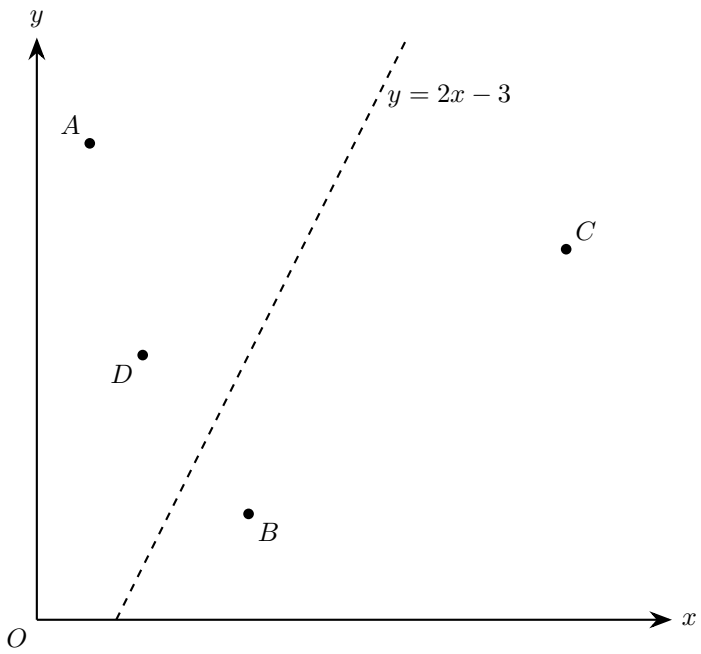
$$M = (7+3)m = 10m$$

For the y-coordinate, with $\bar{y} = k$:

$$Mk = m(0) + 2m(3) + 4m(-1) + 3m(2)$$

So

$$\begin{aligned} 10mk &= 0 + 6m - 4m + 6m \\ 10mk &= 8m \\ k &= \frac{8}{10} \\ k &= \frac{4}{5} \end{aligned}$$



5. The diagram shows four particles, A, B, C and D, fixed in a horizontal plane which contains the x- and y-axes.

Particle A has mass 3 kg and is attached at the point $(1, 9)$.
Particle B has mass 4 kg and is attached at the point $(4, 2)$.
Particle C has mass 5 kg and is attached at the point $(10, 7)$.
Particle D has mass m kg and is attached at the point $(2, 5)$.

The centre of mass of the four particles lies on the line $y = 2x - 3$.
Find the value of m and the coordinates of the centre of mass. **[6]**

Solution

Let the centre of mass be $(\bar{x}, \bar{y})$.

For particles of masses m_i at the points (x_i, y_i), the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = 3 + 4 + 5 + m = (12 + m)\,\text{kg}$$

For the x-coordinate:

$$M\bar{x} = 3(1) + 4(4) + 5(10) + m(2)$$

so

$$\begin{aligned}(12 + m)\bar{x} &= 3 + 16 + 50 + 2m \\ &= 69 + 2m\end{aligned}$$

Hence

$$\bar{x} = \frac{69 + 2m}{12 + m}$$

For the y-coordinate:

$$M\bar{y} = 3(9) + 4(2) + 5(7) + m(5)$$

so

$$\begin{aligned}(12+m)\bar{y} &= 27 + 8 + 35 + 5m \\ &= 70 + 5m\end{aligned}$$

Hence

$$\bar{y} = \frac{70+5m}{12+m}$$

The centre of mass lies on the line $y = 2x - 3$, so

$$\frac{70+5m}{12+m} = 2\left(\frac{69+2m}{12+m}\right) - 3$$

Multiply through by $12 + m$:

$$\begin{aligned}70 + 5m &= 2(69 + 2m) - 3(12 + m) \\ &= 138 + 4m - 36 - 3m \\ &= 102 + m\end{aligned}$$

Therefore

$$\begin{aligned}70 + 5m &= 102 + m \\ 4m &= 32 \\ m &= 8\end{aligned}$$

Now substitute $m = 8$. Then the total mass is

$$M = 12 + 8 = 20\,\text{kg}$$

So

$$\begin{aligned}\bar{x} &= \frac{69 + 2(8)}{20} = \frac{85}{20} = \frac{17}{4} \\ \bar{y} &= \frac{70 + 5(8)}{20} = \frac{110}{20} = \frac{11}{2}\end{aligned}$$

Therefore

$$m = 8$$

and the centre of mass is

$$\left(\frac{17}{4}, \frac{11}{2}\right)$$

6. Four particles of masses m, $2m$, $3m$ and km are positioned at the points with coordinates $(-a, a)$, $(2a, -2a)$, $(0, 0)$ and $(\mu a, 2\mu a)$ respectively, where k and μ are constants.

The centre of mass of the four particles is at the point with coordinates $(2a, 3a)$.

(a) Find the value of k. **[6]**

(b) Find the value of μ. **[3]**

Solution

(a) For particles of masses m_i at the points (x_i, y_i), the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = m + 2m + 3m + km = (6 + k)m$$

For the x-coordinate, with $\bar{x} = 2a$:

$$M(2a) = m(-a) + 2m(2a) + 3m(0) + km(\mu a)$$

so

$$\begin{aligned}(6 + k)m(2a) &= -am + 4am + 0 + k\mu am \\ 2(6 + k) &= 3 + k\mu \\ 12 + 2k &= 3 + k\mu \\ k\mu &= 9 + 2k\end{aligned}$$

For the y-coordinate, with $\bar{y} = 3a$:

$$M(3a) = m(a) + 2m(-2a) + 3m(0) + km(2\mu a)$$

so

$$\begin{aligned}(6 + k)m(3a) &= am - 4am + 0 + 2k\mu am \\ 3(6 + k) &= -3 + 2k\mu \\ 18 + 3k &= -3 + 2k\mu \\ 2k\mu &= 21 + 3k\end{aligned}$$

Now double the equation $k\mu = 9 + 2k$:

$$2k\mu = 18 + 4k$$

Equating the two expressions for $2k\mu$:

$$\begin{aligned}18 + 4k &= 21 + 3k \\ k &= 3\end{aligned}$$

(b) Substitute $k = 3$ into

$$k\mu = 9 + 2k$$

to get

$$3\mu = 9 + 2(3)$$

$$3\mu = 15$$

$$\mu = 5$$

7. Three particles of masses m, $2m$ and km are placed at the points $A(a, 0)$, $B(-2a, 3a)$ and $C(2a, -a)$ respectively, where $k > 0$.

The centre of mass of the three particles is the point with coordinates $(\bar{x}, \bar{y})$.

(a) (i) Find $\bar{x}$ in terms of a and k.
(ii) Find $\bar{y}$ in terms of a and k. **[4]**

(b) Given that the centre of mass lies on the circle with centre $(a, 0)$ and radius $\frac{\sqrt{2}}{2}a$, find the possible values of k. **[4]**

Solution

(a) For particles of masses m_i at the points (x_i, y_i), the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = m + 2m + km = (k + 3)m$$

(i) For the x-coordinate:

$$M\bar{x} = m(a) + 2m(-2a) + km(2a)$$

So

$$\begin{aligned}(k + 3)m\bar{x} &= am - 4am + 2kam \\ &= am(2k - 3)\end{aligned}$$

Hence

$$\bar{x} = \frac{a(2k - 3)}{k + 3}$$

(ii) For the y-coordinate:

$$M\bar{y} = m(0) + 2m(3a) + km(-a)$$

So

$$\begin{aligned}(k + 3)m\bar{y} &= 0 + 6am - kam \\ &= am(6 - k)\end{aligned}$$

Hence

$$\bar{y} = \frac{a(6 - k)}{k + 3}$$

(b) The centre of mass lies on the circle with centre $(a, 0)$ and radius $\frac{\sqrt{2}}{2}a$, so

$$(\bar{x} - a)^2 + \bar{y}^2 = \left(\frac{\sqrt{2}}{2}a\right)^2 = \frac{a^2}{2}$$

Using part (a):

$$\begin{aligned}\bar{x} - a &= \frac{a(2k-3)}{k+3} - a \\ &= a\left(\frac{2k-3-(k+3)}{k+3}\right) \\ &= \frac{a(k-6)}{k+3}\end{aligned}$$

and

$$\bar{y} = \frac{a(6-k)}{k+3} = -\frac{a(k-6)}{k+3}$$

Substituting into the circle equation:

$$\left(\frac{a(k-6)}{k+3}\right)^2 + \left(-\frac{a(k-6)}{k+3}\right)^2 = \frac{a^2}{2}$$

$$2\left(\frac{a(k-6)}{k+3}\right)^2 = \frac{a^2}{2}$$

So

$$4(k-6)^2 = (k+3)^2$$

Expanding:

$$4(k^2 - 12k + 36) = k^2 + 6k + 9$$

$$4k^2 - 48k + 144 = k^2 + 6k + 9$$

$$3k^2 - 54k + 135 = 0$$

$$k^2 - 18k + 45 = 0$$

Factorising:

$$(k-3)(k-15) = 0$$

Hence the possible values of k are

$$k = 3 \quad \text{or} \quad k = 15$$

8. Three particles of masses $2m$, $5m$ and $3m$ are placed at the points $(1, -4)$, $(-3, 2)$ and $(q, 2q - 1)$ respectively.

The value of q is such that the distance of the centre of mass of the three particles from the point $(1, 0)$ is as small as possible.

Find the value of q. **[7]**

Solution

Let the centre of mass be $G(\bar{x}, \bar{y})$.

For particles of masses m_i at the points (x_i, y_i), the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = 2m + 5m + 3m = 10m$$

For the x-coordinate:

$$10m\bar{x} = 2m(1) + 5m(-3) + 3m(q)$$

so

$$10\bar{x} = 2 - 15 + 3q$$

$$\bar{x} = \frac{3q - 13}{10}$$

For the y-coordinate:

$$10m\bar{y} = 2m(-4) + 5m(2) + 3m(2q - 1)$$

so

$$10\bar{y} = -8 + 10 + 6q - 3$$

$$= 6q - 1$$

$$\bar{y} = \frac{6q - 1}{10}$$

Hence

$$G\left(\frac{3q - 13}{10}, \frac{6q - 1}{10}\right)$$

Let the distance from G to $(1, 0)$ be d. Then

$$d^2 = \left(\frac{3q - 13}{10} - 1\right)^2 + \left(\frac{6q - 1}{10}\right)^2$$

$$= \left(\frac{3q - 23}{10}\right)^2 + \left(\frac{6q - 1}{10}\right)^2$$

$$= \frac{(3q - 23)^2 + (6q - 1)^2}{100}$$

Now expand:

$$(3q-23)^2+(6q-1)^2 = 9q^2-138q+529+36q^2-12q+1$$
$$= 45q^2-150q+530$$

So

$$d^2 = \frac{45q^2-150q+530}{100}$$

To make d as small as possible, it is enough to make d^2 as small as possible. Complete the square:

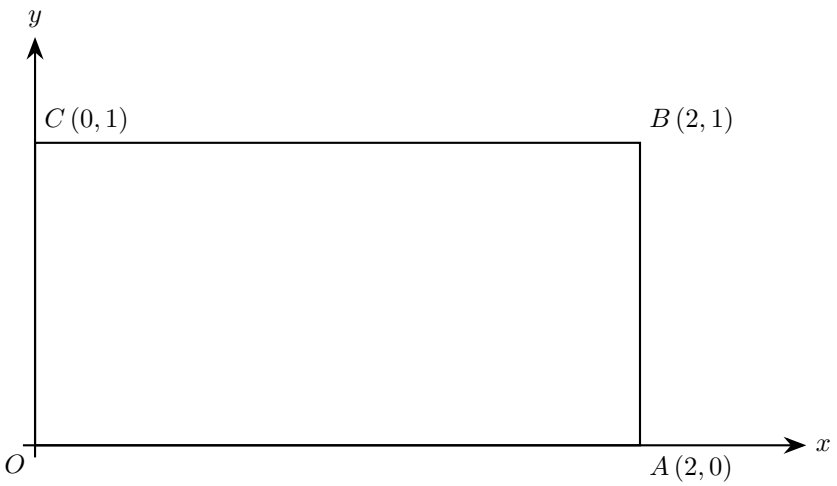
$$45q^2-150q+530 = 45\left(q^2-\frac{10}{3}q\right)+530$$
$$= 45\left[\left(q-\frac{5}{3}\right)^2-\frac{25}{9}\right]+530$$
$$= 45\left(q-\frac{5}{3}\right)^2-125+530$$
$$= 45\left(q-\frac{5}{3}\right)^2+405$$

Since $45\left(q-\frac{5}{3}\right)^2 \geq 0$, this expression is smallest when

$$q-\frac{5}{3}=0$$

Therefore

$$q=\frac{5}{3}$$



9. The diagram shows a light rectangular lamina $OABC$ where O is the origin of the coordinate system in which the points A, B and C have coordinates $(2, 0)$, $(2, 1)$ and $(0, 1)$ respectively. Coordinates refer to the axes shown and the units are metres.

The following three forces are applied to the lamina.

$\begin{pmatrix} p \\ 2q \end{pmatrix}$ N at the point $(1, 0)$

$\begin{pmatrix} -q \\ p \end{pmatrix}$ N at the point C

$\begin{pmatrix} -3 \\ -6 \end{pmatrix}$ N at the point $(t, 1)$, where $0 \leq t \leq 2$

The three forces form a couple of magnitude 3 N m in the anticlockwise direction.

(a) Determine the values of p, q and t. **[5]**

The three forces are removed. The lamina is in the same position as shown in the diagram.

Three particles are now attached to certain points on the lamina as follows.

(b) Find the coordinates of the composite body's centre of mass. Give your answer in terms of λ, where the particles are

- a particle of mass 2 kg at the point $(2\lambda^2, \lambda)$, where $0 \leq \lambda \leq 1$
- a particle of mass 1 kg at the point A
- a particle of mass 1 kg at the point C

[4]

(c) Show that, as λ varies, the centre of mass of the combined object lies on the curve

$$x = 4y^2 - 2y + k$$

where k is a constant to be determined. **[3]**

Solution

(a) Since the three forces form a couple, their resultant force is zero.

Resolving in the x- and y-directions:

$$p - q - 3 = 0, \qquad p + 2q - 6 = 0$$

Subtracting the first equation from the second gives

$$3q - 3 = 0$$

so

$$q = 1$$

Substituting into $p - q - 3 = 0$:

$$p - 1 - 3 = 0 \quad \Longrightarrow \quad p = 4$$

Now take moments about O, with anticlockwise positive, treating the two components of each force separately.

For the force at $(1, 0)$, the x-component acts along the x-axis, through O. The y-component, $2q$ N in the positive y-direction, acts at a perpendicular distance of 1 m from O, so its moment is anticlockwise.

For the force at C, the y-component acts along the y-axis, through O. The x-component, q N in the negative x-direction, acts at a perpendicular distance of 1 m from O, so its moment is anticlockwise.

For the force at $(t, 1)$, the x-component, 3 N in the negative x-direction, acts at a perpendicular distance of 1 m from O, so its moment is anticlockwise. The y-component, 6 N in the negative y-direction, acts at a perpendicular distance of t m from O, so its moment is clockwise.

The forces form an anticlockwise couple of magnitude 3 N m, so

$$2q(1) + q(1) + 3(1) - 6t = 3$$

Using $q = 1$:

$$2 + 1 + 3 - 6t = 3$$

Therefore

$$6 - 6t = 3 \quad \Longrightarrow \quad 6t = 3 \quad \Longrightarrow \quad t = \frac{1}{2}$$

So

$$p = 4, \qquad q = 1, \qquad t = \frac{1}{2}$$

(b) The lamina is light, so its mass is taken to be zero.

Let the centre of mass of the composite body be $(\bar{x}, \bar{y})$.

For particles of masses m_i at the points (x_i, y_i), the centre of mass is given by

$$M\bar{x} = \sum m_i x_i, \qquad M\bar{y} = \sum m_i y_i$$

where $M = \sum m_i$ is the total mass, which here is

$$M = 2 + 1 + 1 = 4 \text{ kg}$$

For the x-coordinate:

$$4\bar{x} = 2(2\lambda^2) + 1(2) + 1(0) = 4\lambda^2 + 2$$

So

$$\bar{x} = \lambda^2 + \frac{1}{2}$$

For the y-coordinate:

$$4\bar{y} = 2(\lambda) + 1(0) + 1(1) = 2\lambda + 1$$

So

$$\bar{y} = \frac{2\lambda + 1}{4}$$

Hence the centre of mass is

$$\left(\lambda^2 + \frac{1}{2}, \frac{2\lambda + 1}{4}\right)$$

(c) From part (b), the centre of mass has coordinates

$$\bar{x} = \lambda^2 + \frac{1}{2}, \qquad \bar{y} = \frac{2\lambda + 1}{4}$$

Rearranging the expression for $\bar{y}$:

$$4\bar{y} = 2\lambda + 1$$

so

$$\lambda = \frac{4\bar{y} - 1}{2}$$

Substitute this into the expression for $\bar{x}$:

$$\begin{aligned}
\bar{x} &= \left(\frac{4\bar{y} - 1}{2}\right)^2 + \frac{1}{2} \\
&= \frac{16\bar{y}^2 - 8\bar{y} + 1}{4} + \frac{1}{2} \\
&= 4\bar{y}^2 - 2\bar{y} + \frac{1}{4} + \frac{1}{2} \\
&= 4\bar{y}^2 - 2\bar{y} + \frac{3}{4}
\end{aligned}$$

Therefore, as λ varies, the centre of mass lies on the curve

$$x = 4y^2 - 2y + k$$

with

$$k = \frac{3}{4}$$